

PricePilot AI

Title: PricePilot AI: Dynamic Pricing Optimization & Revenue Intelligence System

Milestone 1 Documentation

Requirements & Dataset Preparation

Milestone 1 (EDA, Frontend & Backend Setup, Git and GitHub Setup)

Project Objective

The objective of **PricePilot AI** is to build an AI-powered dynamic pricing platform that helps businesses optimize product prices based on market demand, competitor pricing, customer behaviour, and sales performance.

The system is designed to support:

- Price prediction
- Competitor pricing analysis
- Demand-based pricing
- Revenue optimization
- Promotion and discount analysis
- Pricing analytics

The platform aims to maximize revenue, improve profitability, enhance market competitiveness, and support intelligent pricing decisions using data analytics and Machine Learning.

This solution can potentially be applied to:

- E-commerce companies
 - Retail businesses
 - Online marketplaces
 - Hotels
 - Sales organizations
-

Dataset Details [Amazon Dataset Link](#)

- **Total Records:** 76,000
- **Total Columns:** 21

The dataset contains information related to retail operations, including sales, inventory, demand, pricing, promotions, competitor pricing, weather conditions, and other external business factors.

Categorical Features

The dataset includes categorical variables such as:

- Store ID
 - Product ID
 - Category
 - Region
 - Weather Condition
 - Seasonality
-

Numerical Features

The dataset contains numerical variables including:

- Inventory Level
- Units Sold
- Units Ordered
- Price
- Discount
- Promotion
- Competitor Pricing
- Epidemic
- Demand

- Revenue

Time Features

The dataset contains date-related information from which the following features can be extracted:

- Year
- Month
- Day

These features can be useful for analysing sales trends and seasonal patterns.

Target Variable

The target variable depends on the specific Machine Learning objective.

Potential target variables include:

- **Revenue**
- **Units Sold**
- **Demand**

Process Followed

Step 1: Environment Setup

The development environment was prepared for data analysis and Machine Learning implementation.

The required Python libraries were imported:

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

Additional Machine Learning libraries can be used in future milestones.

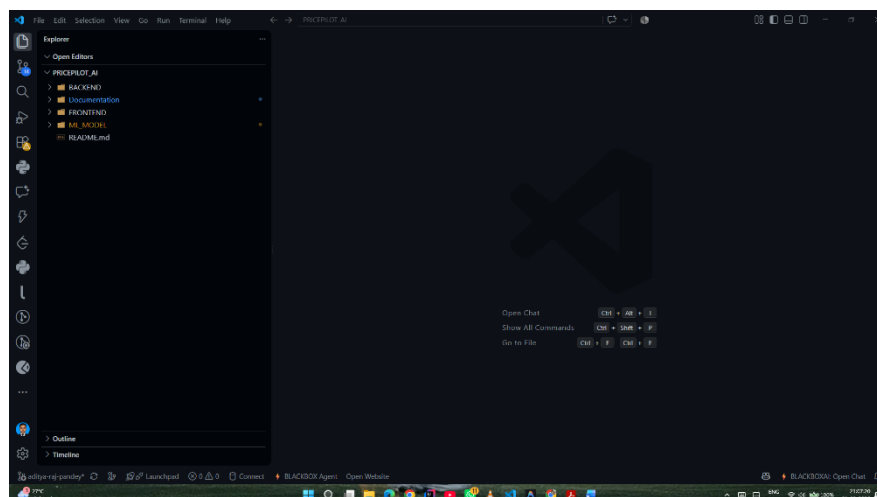
The following tools and technologies were used:

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

1. Creating Python Environment & Activation

1. `python -m venv venv`
2. `venv\Scripts\activate`

2. Frontend & Backend Setup



Step 2: Data Exploration

```
SALES DATA --> kaggle

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
print("✅ All dependencies successfully imported!")

import kagglehub
path = kagglehub.dataset_download("karkavelrajaj/amazon-sales-dataset")

Using Colab cache for faster access to the 'amazon-sales-dataset' dataset.
```

```
Data Inspection

print("Shape:", df.shape)

print("\nColumns:")
print(df.columns.tolist())

print("\nDataset Info:")
df.info()

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Rows:")
print(df.duplicated().sum())

... Shape: (76000, 21)

Columns:
['Date', 'Store ID', 'Product ID', 'Category', 'Region', 'Inventory Level', 'Units Sold', 'Units Ordered', 'Price', 'Discount', 'Weat
```

Observations

The following observations were made during initial data exploration:

- The dataset contains **76,000 records**.
- The dataset contains **21 columns**.
- The dataset contains both numerical and categorical variables.
- The dataset includes important business variables such as Revenue, Demand, Price, Units Sold, Discount, and Competitor Pricing.
- The dataset is suitable for detailed Exploratory Data Analysis.
- Date-related features can be used for temporal and seasonal analysis.

Results

The analysis showed:

- **No missing values**
- **No duplicate records**

Therefore, no records needed to be removed because of missing or duplicate data.

Step 4: Outlier Analysis

Outliers were analyzed using statistical methods such as the **Interquartile Range (IQR)**.

The IQR method uses:

$$\text{IQR} = Q3 - Q1$$

The lower and upper limits are calculated as:

$$\text{Lower Bound} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper Bound} = Q3 + 1.5 \times \text{IQR}$$

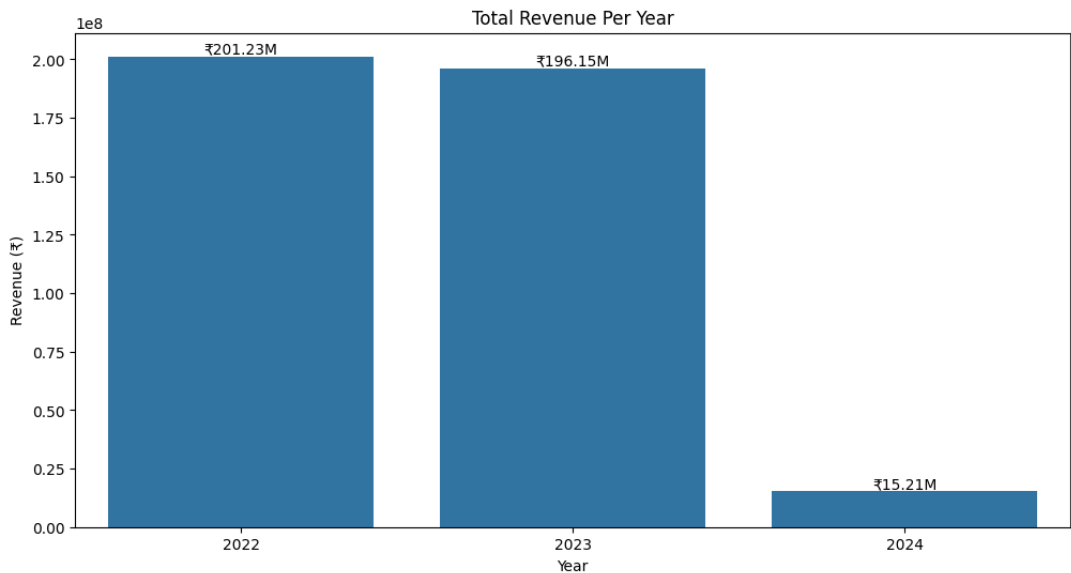
Outliers were observed in some numerical variables, particularly:

- Discount
- Units Ordered
- Inventory Level
- Revenue

Instead of removing a large number of records, selected outliers were handled carefully to preserve the size and information contained in the dataset.

Data Analysis (Graphs)

Graph 1: Distribution of Revenue



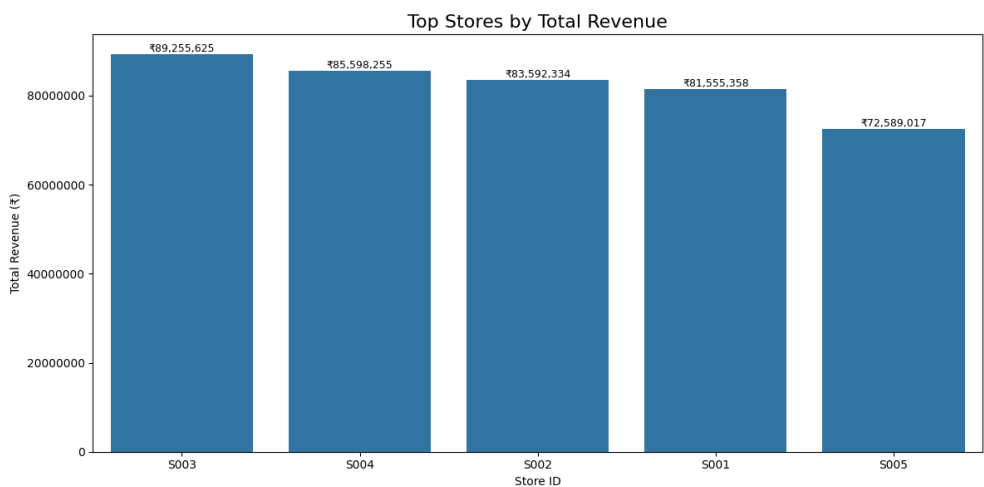
Observation

The Revenue distribution helps identify:

- The concentration of revenue values
- High and low revenue observations
- Possible skewness
- Potential outliers

This analysis provides an initial understanding of overall business performance.

Graph 2: Top Stores By revenue



Observations

The correlation analysis revealed several important relationships.

Demand vs Units Sold

Correlation:

0.83

This represents a strong positive relationship.

Higher demand is strongly associated with higher sales.

Price vs Competitor Pricing

Correlation:

0.98

This represents an extremely strong positive relationship.

This suggests that product prices closely move with competitor pricing.

Price vs Revenue

Correlation:

0.72

This indicates a strong positive relationship between product price and revenue.

Promotion vs Discount

Correlation:

0.78

Promotions are strongly associated with discounts.

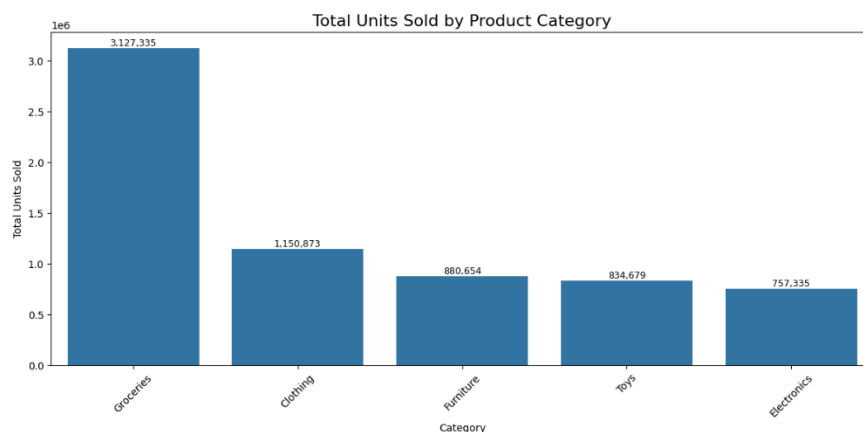
Epidemic Impact

Negative correlations were observed between epidemic conditions and:

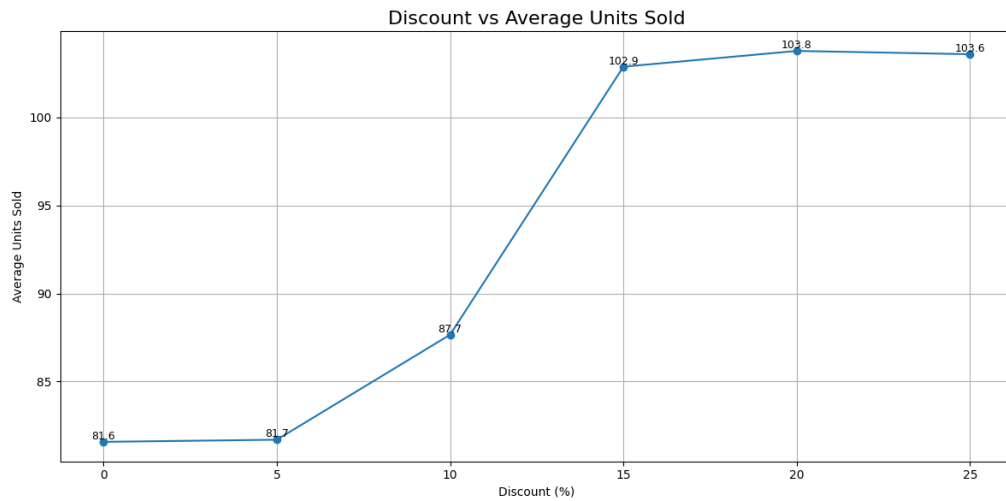
- Demand
- Units Sold
- Revenue

This suggests that epidemic conditions may negatively affect business performance.

Graph 3: Product category Distribution



Graph:4 Discounts Analysis on Sales



Observation

The analysis showed that:

- Low discounts have limited impact on sales.
- Sales increase significantly between 10% and 15% discounts.
- Average units sold reaches its highest level around 20% discount.
- Increasing the discount beyond 20% does not significantly increase sales.

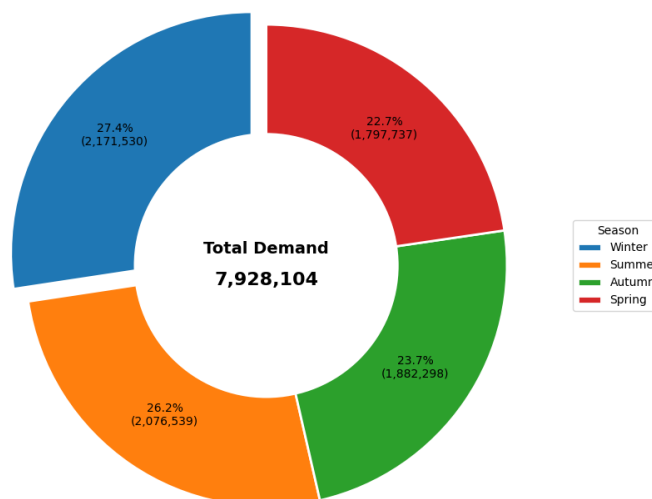
Conclusion

This indicates **diminishing returns** from excessive discounts.

A discount range of approximately **15–20%** appears effective for increasing sales volume.

Graph 5: Demand According to seasons

Demand Distribution by Seasonality



Feature Engineering

Date Feature Extraction

Date-related features can be extracted from the dataset.

```
df['Date'] = pd.to_datetime(df['Date'])
```

```
df['Year'] = df['Date'].dt.year
```

```
df['Month'] = df['Date'].dt.month
```

```
df['Day'] = df['Date'].dt.day
```

These features can help analyze:

- Monthly sales
- Yearly revenue
- Seasonal demand

- Business trends
-

Potential Feature Engineering

A useful feature for pricing analysis is:

Price Difference

```
df['Price_Difference'] = (  
    df['Price'] -  
    df['Competitor Pricing']  
)
```

This feature can help analyze:

- Whether the company is more expensive than competitors
 - Whether lower prices increase sales
 - The effect of competitive pricing on revenue
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Challenges Faced

The following challenges were encountered during Milestone 1:

1. Understanding the Dataset and VS code setup
 2. Outlier Detection
 3. Interpretation of Correlation
 4. Different Metric Scales
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Conclusion

Milestone 1 successfully completed the **dataset exploration, cleaning, and Exploratory Data Analysis phase** of the **PricePilot AI project**. Also, Frontend and Backend setup also Done.

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